Global, regional, and national burden of male breast cancer in 204 countries and territories: a systematic analysis from the global burden of disease study, 1990–2021

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Summary

Background Male breast cancer (MBC) is a rare malignancy that has been under-investigated, with limited global epidemiological research dedicated to it. A comprehensive estimate of the global, regional, and national burden of MBC is valuable for policy planning. This study aims to evaluate the burden of MBC across 204 countries and territories.

Methods MBC data were collected from the 2021 Global Burden of Diseases, Injuries, and Risk Factors (GBD) Study estimates spanning from 1990 to 2021. The global incidence, prevalence, deaths, and disability-adjusted life-years (DALYs) attributed to MBC, as well as corresponding age-standardized rates, were calculated. Temporal trends, projections of incidence and mortality to 2050, lifetime risk, and risk factors of MBC were also estimated according to regions and countries.

Findings In 2021, there were 38,827 (95% uncertainty interval [UI], 24,650–47,846) new cases, 320,459 (95% UI, 220,533–384,317) prevalent cases, 13,274 (95% UI, 9074–16,240) deaths, and 380,917 (95% UI, 252,922–476,417) DALYs attributed to MBC worldwide, with the highest disease burden observed in the Eastern Sub-Saharan Africa region. From 1990 to 2021, the age-standardized incidence and mortality rates of MBC significantly increased, but they are projected to decrease over the next 30 years. High-middle Socio-demographic Index (SDI) quintile had the highest lifetime risk of developing MBC, while the low SDI quintile had the highest lifetime risk of dying from MBC. Dietary risk and alcohol use were identified as important risk factors for MBC deaths and DALYs globally.

Interpretation The global burden of MBC significantly increased from 1990 to 2021, with notable geographic disparities. Efforts aimed at MBC prevention and control strategies should take into account the inequities in its global distribution.

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Keywords: Male breast cancer; Global burden; Epidemiology; Lifetime risk

Introduction

In comparison to the most common form of breast cancer in females, male breast cancer (MBC) is a rare malignancy.[1] It has been reported that MBC accounts for only 0.6%–1% of all breast cancer cases and represents approximately 0.3% of all newly diagnosed cancers in men.[2,3] According to data from the Surveillance, Epidemiology, and End Results program, the age-adjusted incidence of invasive MBC was 1.11 per 100,000 from 1975 to 2015 in the United States.[4] Typically, MBC patients are diagnosed at an older age, with lower-grade, later clinicopathological stage, and more estrogen receptor-positive tumors than females.[5,6] Due to the limited study on the etiology and few randomized

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Research in context

Evidence before this study

We systematically searched PubMed, Web of Science, and Embase up to December 2024 using keywords related to male breast cancer (MBC), including "male breast cancer", "breast cancer", "incidence', "deaths", "prevalence", "disability-adjusted life-years", "global burden", "epidemiology", and "trend analysis". Previous research on MBC is limited due to its rarity, with most studies being region-specific or based on outdated data, and global studies have been scarce. The contribution of MBC to overall cancer burden, particularly regarding temporal trends, risk factors, and geographic disparities, remains underexplored.

Added value of this study

This study is the first to provide a comprehensive, global analysis of MBC using data from the Global Burden of Disease Study 2021, covering 204 countries and territories from 1990 to 2021. It identifies geographic disparities and explores the influence of risk factors like diet, alcohol use, and tobacco consumption on MBC-related deaths and disability-adjusted life-years (DALYs), offering new insights into the epidemiology of MBC. Moreover, this study provides lifetime risk estimates and projections of MBC incidence, prevalence, mortality, and DALYs to 2050, filling a critical gap in the literature and guiding healthcare strategies for MBC prevention and control.

Implications of all the available evidence

The evidence indicates that MBC poses a significant global public health challenge, with notable geographic disparities in its burden. Although global incidence and mortality rates have increased from 1990 to 2021, they are expected to decrease by 2050; however, regions such as Oceania and Sub-Saharan Africa may see rising rates due to demographic changes and shifting risk factors by 2025. This highlights the need for interventions addressing modifiable risk factors like diet, alcohol, and tobacco use, particularly in high-risk regions, while also emphasizing the importance of public awareness and tailored prevention strategies to reduce the global burden of MBC.

clinical trials for MBC, the management of male patients with breast cancer is generalized from the treatment standards of female breast cancer.[7] However, male patients with breast cancer generally experience poorer overall survival compared to female patients.[8] Although overall survival rates have improved over time for both male and female breast cancer, less improvement has been achieved for men.[1]

Due to its rarity, MBC has been neglected and under-investigated, with a lack of global epidemiological research. Although some studies have evaluated the incidence of MBC, they were limited to certain countries or regions, or they reported relatively outdated data.[2,9–11] Therefore, to enhance awareness of MBC, this study comprehensively estimated the global disease burden, temporal trends, predictions, risk factors, and lifetime risk of MBC using high-quality international and national cancer registries. To the best of our knowledge, this is the first comprehensive study providing an in-depth overview of the global epidemiological landscape and temporal trends of MBC. Healthcare policymakers, funding agencies, and researchers can develop strategies to address the burden of MBC based on these data, especially in high-risk locations.

Methods

Data sources

This study adheres to the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) checklist and Guidelines for Accurate and Transparent Health Estimates Reporting (GATHER) checklist. We have documented the steps in our analytical procedures and detailed the data sources used. The analysis in this study was based on Global Burden of Diseases, Injuries, and Risk Factors (GBD) 2021 estimates, which were released in 2024. These estimates provide an epidemiological assessment of 371 diseases and injuries, including incidence, prevalence, mortality, years lived with disability (YLDs), years of life lost (YLLs), and disability-adjusted life-years (DALYs), for 204 countries and territories from 1990 to 2021.[12] Data sources for this study were accessed using the Global Health Data Exchange (GHDx) online query tool (http://ghdx.healthdata.org/gbd-results-tool). GBD 2021 is the most recent iteration, and detailed descriptions have been reported previously.[12,13] The GBD 2021, with the support of more than 11,500 contributors from 164 countries, provides a comprehensive evaluation of global health and disease burden through detailed data collection, review, and analysis.[14] The GBD data collection involves multiple sources, including epidemiological surveys, hospital records, vital registration systems, disease monitoring systems, as well as academic papers and policy reports (https://ghdx.healthdata.org/gbd-2021/sources). These data are standardized using International Classification of Diseases (ICD) codes to ensure accuracy and comparability. Complex modeling tools, such as DisMod-MR and spatiotemporal Gaussian process regression (ST-GPR), are used to estimate the burden of cancer.[12] Data processing includes adjustments for heterogeneity and bias, as well as uncertainty analysis through Monte Carlo simulations. Details of the disease model of the MBC are presented in the GBD 2021 methods appendices (https://www.healthdata.org/gbd/methods-appendices-2021/cancers).

Measures and definitions

Four measures of MBC disease burden were used in this study: incidence, prevalence, deaths, and DALYs. Incidence refers to the number of new cases during a given period in the male population. In the GHDx tool, incidence is expressed as the number of new cases in a year divided by the mid-year population size. Prevalence is defined as the proportion of people who are MBC cases in a population. All prevalence results in the GBD are reported as point prevalence rates, which reflect the number of MBC cases at a particular time divided by the population at that time. Deaths refer to the number of deaths occurring in MBC patients during a certain time period. DALYs are defined as years of healthy life lost due to MBC; one DALY represents one year of lost full health. We did not include YLDs and YLLs as separate measures because DALYs is the sum of years lost due to YLDs and YLLs. Age-standardized rates were used for all measures to adjust for variations in age distribution across different populations, based on the GBD world population standard for each age group (Appendix 1),[15] facilitating the comparison of disease burdens across diverse populations.

Geographic units and socio-demographic index

GBD 2021 estimated epidemiological measures for 204 countries and territories that were grouped into seven super regions including 21 regions based on their epidemiologic similarity and geographic proximity. The seven super regions are: (1) Central Europe, Eastern Europe, and Central Asia; (2) high-income regions including Australasia, high-income Asia Pacific, high-income North America, Southern Latin America, and Western Europe; (3) Latin America and the Caribbean regions including Andean Latin America, Caribbean, Central Latin America, and Tropical Latin America; (4) North Africa and the Middle East; (5) South Asia; (6) Southeast Asia, East Asia, and Oceania; and (7) Sub-Saharan Africa, including Central Sub-Saharan Africa, Eastern Sub-Saharan Africa, Southern Sub-Saharan Africa, and Western Sub-Saharan Africa.[16] The Socio-demographic Index (SDI) is a composite development indicator for a country, incorporating lag-distributed income per capita, average years of schooling, and the total fertility rate in females under the age of 25 years.[17] The 204 countries and territories were divided into five SDI quintiles based on their SDI levels: low, low-middle, middle, high-middle, and high SDI.[18]

Risk factors

The GBD 2021 includes a total of 88 risk factors, categorized as environmental/occupational, behavioral, and metabolic risks, following a hierarchy of risks in GBD risk assessments.[13] Accordingly, alcohol use, dietary risk, and tobacco are the only risk factors reported for MBC by the GBD 2021 study, based on the risk–outcome pairs assessed as meeting the World Cancer Research Fund criteria of convincing or likely causal evidence. Therefore, MBC deaths and DALYs attributable to alcohol use, dietary risk, and tobacco are reported in this study. The GBD comparative risk assessment framework was used to compute the fraction of cancer-specific burden attributable to each risk factor.[19] In order to assess the influence of these risk factors, we utilized sophisticated approaches, such as DisMod-MR 2.1 and ST-GPR.[13] These methods enabled us to model the distribution of exposures across different populations and regions. Based on epidemiological evidence, we established the theoretical minimum risk exposure level (TMREL) for each factor, which represents the optimal exposure level for minimizing the risk of MBC. By integrating exposure data, relative risk estimates, and TMRELs, we calculated the population attributable fractions (PAFs) for each risk factor. These PAFs were stratified by region and quantified the potential reduction in the MBC burden if exposure to a given risk factor were reduced to its TMREL. Through this approach, we highlighted the contribution of modifiable risk factors. This provided a more nuanced understanding of the MBC burden and potential directions for interventions across different populations. The details of the definitions and quantifying methodology of the risk attribution are described elsewhere.[13]

Statistics

Age-standardized rates (per 100,000 males) with 95% uncertainty intervals (UI) were used to quantify the incidence, prevalence, mortality, and DALYs of MBC by time, region, country, and SDI. GBD calculated the uncertainty intervals for all metrics using the mean estimate across 1000 draws, with 95% UIs represented by the 2.5th and 97.5th percentiles of this distribution. Temporal trends in age-standardized rates of incidence, prevalence, mortality, and DALYs were determined by measuring the average annual percent change (AAPC) and corresponding 95% confidence interval (CI) for the periods 1990–2021, 1990–2000, 2001–2010, and 2011–2021 using joinpoint regression analysis (Joinpoint Trend Analysis Software, NCI, version 5.0, 2023). A positive AAPC indicates an upward trend, and vice versa. An AAPC with a 95% CI including 0 indicates a stable trend without a significant change. Bayesian Age-Period-Cohort (BAPC) Models[20] implemented within the Integrated Nested Laplace Approximations (INLA) framework were used to forecast age-standardized rates of MBC incidence, prevalence, deaths, and DALYs from 2022 to 2050. This approach was chosen for its computational efficiency and reduced error rates compared to traditional methods.[20,21] By combining the INLA framework with the BAPC model, we efficiently approximated the marginal posterior distributions, thereby addressing the mixing and convergence issues often encountered with traditional Bayesian methods that rely on Markov Chain Monte Carlo sampling

techniques. These forecasts were based on Global Population Forecasts 2017–2100 data, utilizing the BAPC and INLA packages available through https://r-forge.r-project.org/ and https://www.r-inla.org/, respectively. The lifetime risk of developing and dying from MBC was calculated using the adjusted for multiple primaries (AMP) method,[22] which incorporates other causes of death and corrects for multiple primary cancers in the incidence rates. MBC incidence and mortality rates, along with all-cause mortality rates, stratified by 5-year age groups, were employed to estimate the lifetime risk of developing or dying from MBC at different ages. These data were all directly sourced from the GBD database. This estimation represents the probability of developing or dying from MBC from a specific age onward. Lifetime is defined as a specific period of time, not necessarily from birth. Thus, when calculating lifetime risk, the cumulative probability is computed from specific points in time, including birth, as well as ages 40, 50, 60, and 70 years, extending until death. A detailed description of lifetime risk methods is provided in the Appendix 2. A P-value less than 0.05 was considered statistically significant. All statistical analyses were performed using R software (version 4.3.2).

Ethics

The study was granted an exemption by the institutional review board, as it utilized publicly available data that did not include any confidential or personally identifiable information.

Role of the funding source

The funders of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Results

Global estimates of MBC burden in 2021

In 2021, there were an estimated 38,827 (95% UI, 24,650–47,846) new cases, 320,459 (95% UI, 220,533–384,317) prevalent cases, 13,274 (95% UI, 9074–16,240) deaths, and 380,917 (95% UI, 252,922–476,417) DALYs of MBC globally. The age-standardized rates per 100,000 people for incidence, prevalence, mortality, and DALYs were 0.94 (95% UI, 0.60–1.15), 7.75 (95% UI, 5.41–9.24), 0.34 (95% UI, 0.23–0.41), and 9.16 (95% UI, 6.12–11.42), respectively, across 204 countries and territories (Supplementary Table S1).

Figs. 1 and 2 show the national-level distribution of age-standardized rates for incidence, prevalence, mortality, and DALYs due to MBC in 2021. The Eastern Sub-Saharan Africa region exhibited the highest age-standardized rates of incidence (3.05 [95% UI, 2.01–5.46]), prevalence (17.86 [95% UI, 12.54–29.95]), mortality (2.39 [95% UI, 1.60–4.29]), and DALYs (57.76 [95% UI, 37.90–105.58]). The high-middle SDI region had the highest age-standardized rates of incidence (1.16 [95% UI, 0.69–1.52]) and prevalence (10.22 [95% UI, 6.62–12.94]), while the low SDI region had the highest age-standardized rates of mortality (0.89 [95% UI, 0.66–1.38]) and DALYs (21.71 [95% UI, 16.08–34.41]) (Supplementary Table S1). Uganda exhibited the highest age-standardized rates of incidence (4.54 [95% UI, 3.03–6.81]), prevalence (26.28 [95% UI, 18.67–37.50]), mortality (3.51 [95% UI, 2.30–5.20]), and DALYs (84.59 [95% UI, 55.88–129.24]) among the 204 countries and territories (Supplementary Table S2).

Globally, the peak incidence (6492 [95% UI, 4060–8145]), prevalence (53,543 [95% UI, 36,239–65,180]), and mortality (2018 [95% UI, 1353–2489]) of MBC occurred in individuals aged 65–69 years, while the highest number of DALYs was in those aged 60–64 years (54,363 [95% UI, 36,273–67,458]). The rates of incidence, prevalence, death, and DALYs increased gradually with age, peaking at age ≥ 95 years for DALYs (55.73 [95% UI, 44.18–64.80]) and mortality (6.56 [95% UI, 5.20–7.63]), with individuals aged 85–89 years exhibiting the highest prevalence (49.23 [95% UI, 43.08–54.91]) and incidence (6.76 [95% UI, 5.50–7.71]) rates (Supplementary Fig. S1).

Global trends of MBC from 1990 to 2021

From 1990 to 2021, the age-standardized rates of incidence, prevalence, mortality, and DALYs for MBC significantly increased, with an AAPC of 1.98 (95% CI, 1.80–2.16) for incidence (rising from 0.52 [95% CI, 0.46–0.60] in 1990 to 0.94 [95% CI, 0.60–1.15] in 2021), 2.03 (95% CI, 1.86–2.19) for prevalence (increasing from 4.26 [95% CI, 3.81–4.76] to 7.75 [95% CI, 5.41–9.24]), 0.61 (95% CI, 0.53–0.70) for mortality (rising from 0.28 [95% CI, 0.24–0.34] to 0.34 [95% CI, 0.23–0.41]), and 0.70 (95% CI, 0.61–0.79) for DALYs (increasing from 7.39 [95% CI, 6.23–8.83] to 9.16 [95% CI, 6.12–11.42]) (Fig. 3A and Supplementary Table S3). Notably, from 1990 to 2000 and 2001 to 2010, the age-standardized rates of incidence, prevalence, mortality, and DALYs showed a significant increasing trend. However, between 2011 and 2021, prevalence continued to rise, incidence increased slightly, while both mortality and DALYs showed a declining trend (Fig. 3B).

Among the 21 geographic regions, East Asia experienced the greatest increase in incidence (AAPC, 4.50 [95% CI, 4.10–4.90]) and prevalence (AAPC, 4.61 [95% CI, 4.26–4.96]) from 1990 to 2021, while Tropical Latin America had the largest increase in mortality and DALYs, with an AAPC of 2.02 (95% CI, 0.76–3.29) and 2.02 (95% CI, 0.71–3.35), respectively (Supplementary Fig. S2 and Table S3). Eastern Europe showed the most significant decline in the age-standardized rates of mortality and DALYs over the period (Supplementary Fig. S2 and Table S3). The middle SDI region saw the highest increase in age-standardized rates of incidence, prevalence, deaths, and DALYs from 1990 to 2021. High

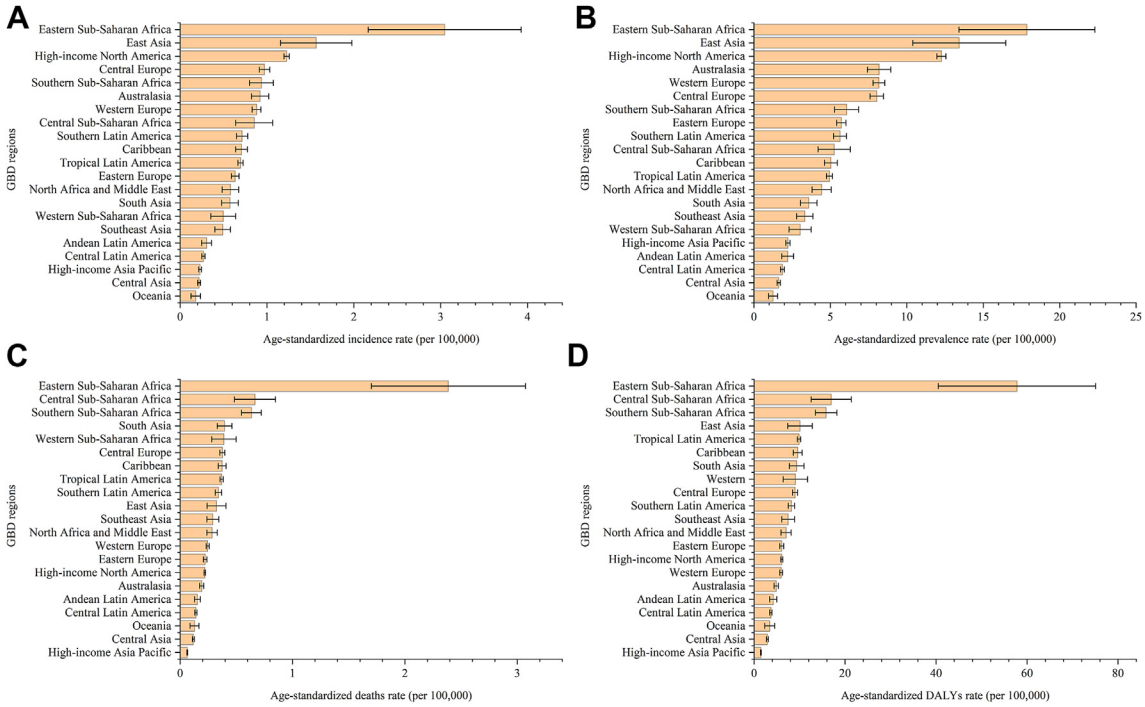


Fig. 1: The age-standardized incidence (A), prevalence (B), death (C), and DALYs (D) rates of male breast cancer in 2021 among 21 GBD regions. Error bars represent the 95% uncertainty intervals (UIs).

SDI and low SDI regions showed significant declines in the age-standardized rates of deaths and DALYs, respectively. Among all 204 countries and territories from 1990 to 2021, Georgia experienced the greatest increase in age-standardized rates of incidence (AAPC, 25.44 [95% CI, 17.99–33.37]), prevalence (AAPC, 7.43 [95% CI, 4.07–10.90]), mortality (AAPC, 25.08 [95% CI, 17.58–33.05]), and DALYs (AAPC, 20.17 [95% CI, 13.28–27.47]) (Supplementary Fig. S2 and Table S4). Conversely, Belarus saw the most significant declines in these rates, with an APCC of –8.14 (95% CI, –9.59–6.66) for incidence, –3.14 (95% CI, –4.85–1.41) for prevalence, –9.54 (95% CI, –11.08–7.79) for mortality, and –9.45 (95% CI, –11.82–7.01) for DALYs.

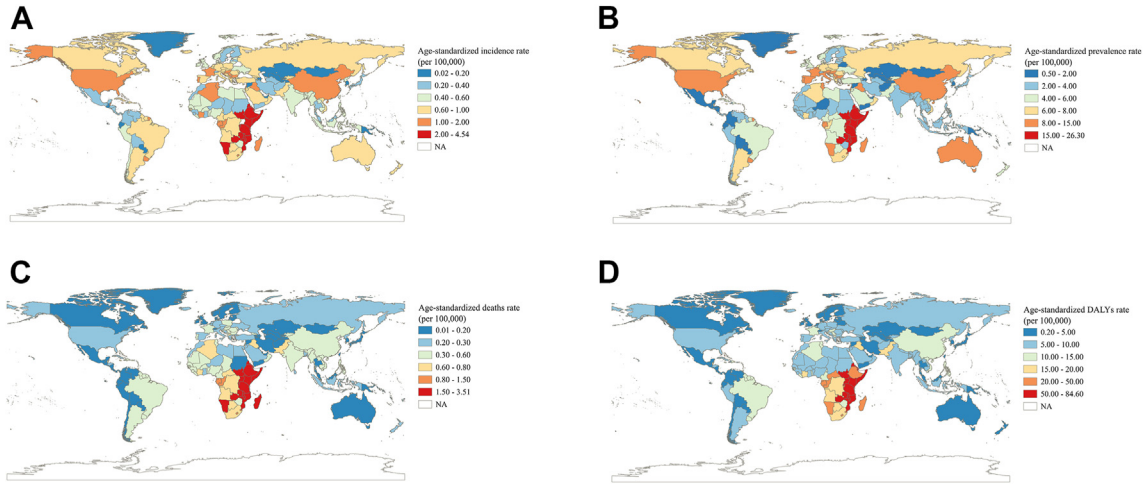
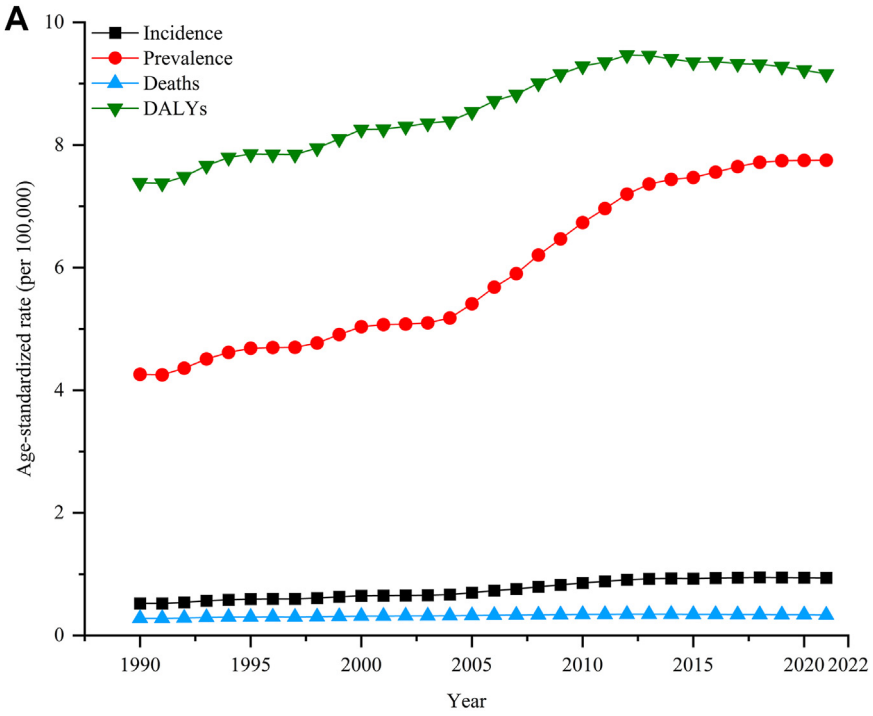


Fig. 2: Age-standardized incidence (A), prevalence (B), death (C), and DALYs (D) rates of male breast cancer per 100,000 population in 2021, by country and territory.



B

	Average Annual Percent Change (AAPC)			
	Incidence	Prevalence	Deaths	DALYs
1990-2021	1.98*	2.03*	0.61*	0.70*
1990-2000	2.08*	1.64*	1.15*	0.97*
2001-2010	3.17*	3.29*	0.90*	1.29*
2011-2021	0.67*	1.14*	-0.25*	-0.20*

Fig. 3: Trends of global age-standardized rates of male breast cancer from 1990 to 2021. (A) Trends from 1990 to 2021. (B) Average annual percent change (AAPC) for the periods 1990–2021, 1990–2000, 2001–2010, and 2011–2021. *The result is statistically significant.

Global lifetime risks of developing and dying from MBC in 2021

From birth, the global lifetime risks of developing MBC was 0.095% (95% CI, 0.064–0.112%) and of dying from MBC was 0.035% (95% CI, 0.025–0.041%) (Fig. 4A and B and Supplementary Table S5). These risks remain relatively stable until the age of 50 years (0.083% [95% CI, 0.058–0.098%] for developing MBC and 0.032% [95% CI, 0.024–0.037%] for death), then gradually decrease with age, with a remaining 0.037% [95% CI, 0.028–0.042%] risk of developing and 0.018% [95% CI, 0.014–0.020%] risk of dying from MBC from the age of 70 years onward (Fig. 4A and B and Supplementary Table S6).

The lifetime risk of developing MBC (0.179% [95% CI, 0.122–0.291%]) and dying from MBC (0.130% [95% CI, 0.09–0.212%]) was highest in Eastern Sub-Saharan Africa (Fig. 4C and D and Supplementary Table S5). In contrast, Oceania had the lowest lifetime risk of developing (0.013% [95% CI, 0.008–0.020%]) and dying from MBC (0.009% [95% CI, 0.006–0.014%]). Among SDI regions, the high-middle SDI region had the highest lifetime risk of developing MBC (0.127% [95% CI, 0.082–0.158%]), whereas the low SDI region had the highest risk of dying from MBC (0.056% [95% CI, 0.044–0.084%]) (Supplementary Table S5). The lifetime risk of developing MBC ranged from 0.002% to 0.336%, and the lifetime risk of dying from MBC ranged from 0 to 0.197% across the 204 countries and territories. People in St. Kitts and Nevis experienced the highest lifetime risk of developing and dying from MBC (Fig. 4C and D and Supplementary Table S7).

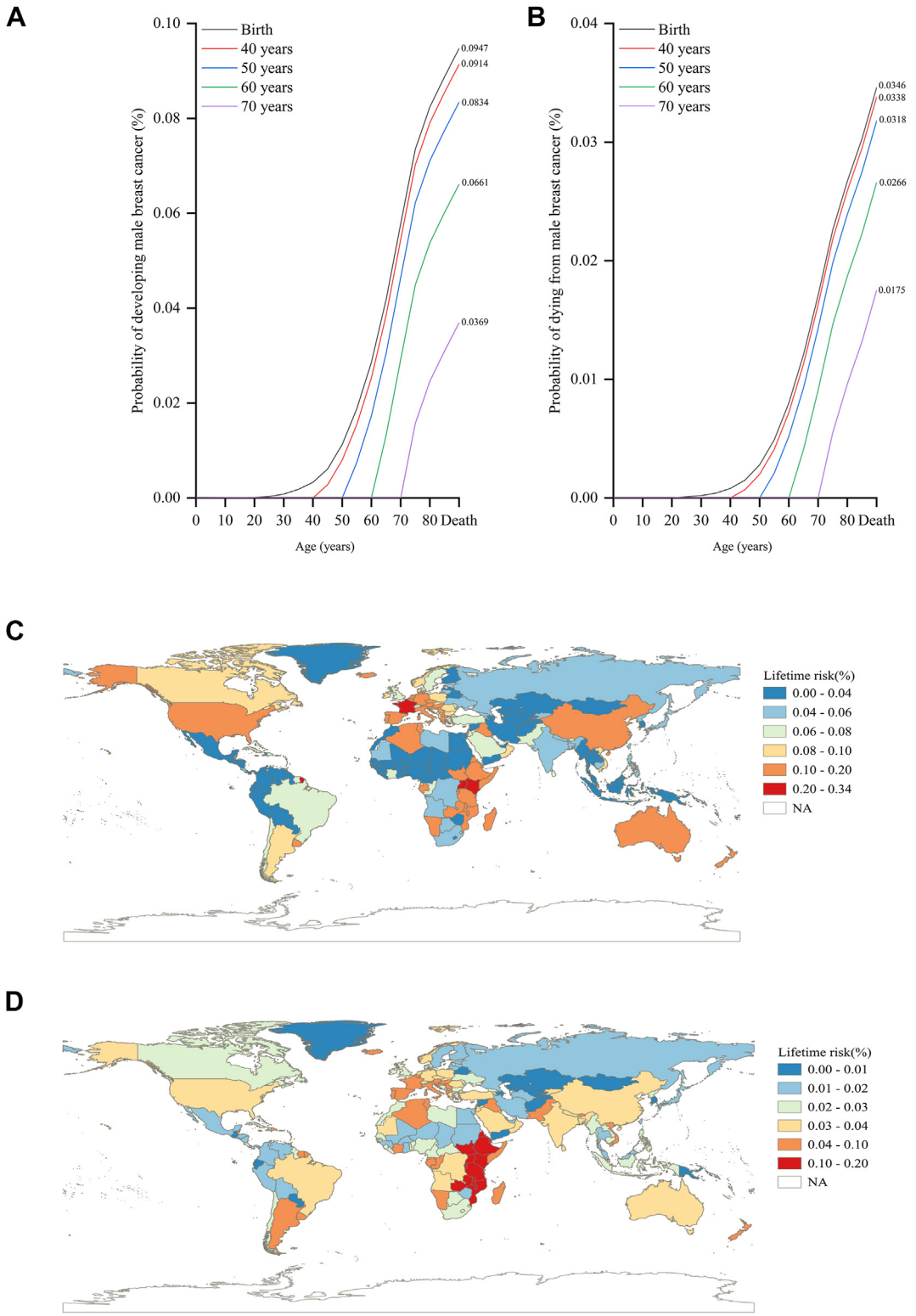


Fig. 4: Global lifetime risks of developing and dying from male breast cancer in 2021. (A) Lifetime risks of developing male breast cancer within selected age intervals. (B) Lifetime risks of dying from male breast cancer within selected age intervals. (C) Lifetime risks of developing male breast cancer from birth by country and territory. (D) Lifetime risks of dying from male breast cancer from birth by country and territory. The curves represent the probabilities of developing or dying from male breast cancer from an age free of male breast cancer to a specific age range.

Global projections of MBC to 2050

Globally, the absolute change in age-standardized incidence rate of MBC is expected to decline by 0.03 between 2022 and 2050 (Fig. 5A). The age-standardized prevalence rate is projected to increase by 0.78, from 7.82 (95% CI, 7.50–8.13) to 8.60 (95% CI, –5.22 to 22.42) per 100,000 males worldwide (Fig. 5B). Conversely, the age-standardized rates of mortality and DALYs are expected to decrease by 0.04 (from 0.33 [95% CI, 0.32–0.34] to 0.29 [95% CI, 0.03–0.55]) and 0.93 (from 9.20 [95% CI, 8.97–9.44] to 8.27 [95% CI, 0.96–15.58]), respectively (Fig. 5C and D).

Regionally, the largest absolute change in age-standardized incidence rate of MBC is observed in Oceania (2.46), followed by Eastern Sub-Saharan Africa (1.18), Southern Latin America (0.93), Central Sub-Saharan Africa (0.74), and Central Latin America (0.41) between 2022 and 2050 (Supplementary Fig. S3 and Table S8). The age-standardized prevalence rate of MBC is predicted to experience the largest increase in Eastern Sub-Saharan Africa (9.49), followed by East Asia (5.96), Central Sub-Saharan Africa (3.93), South Asia (2.76), and North Africa and Middle East (1.49). Oceania (0.86) is projected to see the largest increase in age-standardized deaths. In contrast, the largest absolute decrease in age-standardized incidence (–0.45) and prevalence (–3.77) rates is projected to occur in Western Europe. The most significant decline in age-standardized mortality rate is expected in Southern Sub-Saharan Africa (–0.23), while the largest decrease in age-standardized DALYs rate is anticipated in Southern Latin America (–5.81).

Among the 204 countries and territories, Eq. Guinea (4.34), S. Sudan (21.58), Djibouti (3.55), and S. Sudan (40.21) are projected to experience the highest increases in age-standardized rates of incidence, prevalence, mortality, and DALYs, respectively, between 2022 and 2050 (Supplementary Fig. S3 and Table S9). The age-standardized rate of incidence is projected to undergo the greatest decrease in Djibouti (–1.36), while

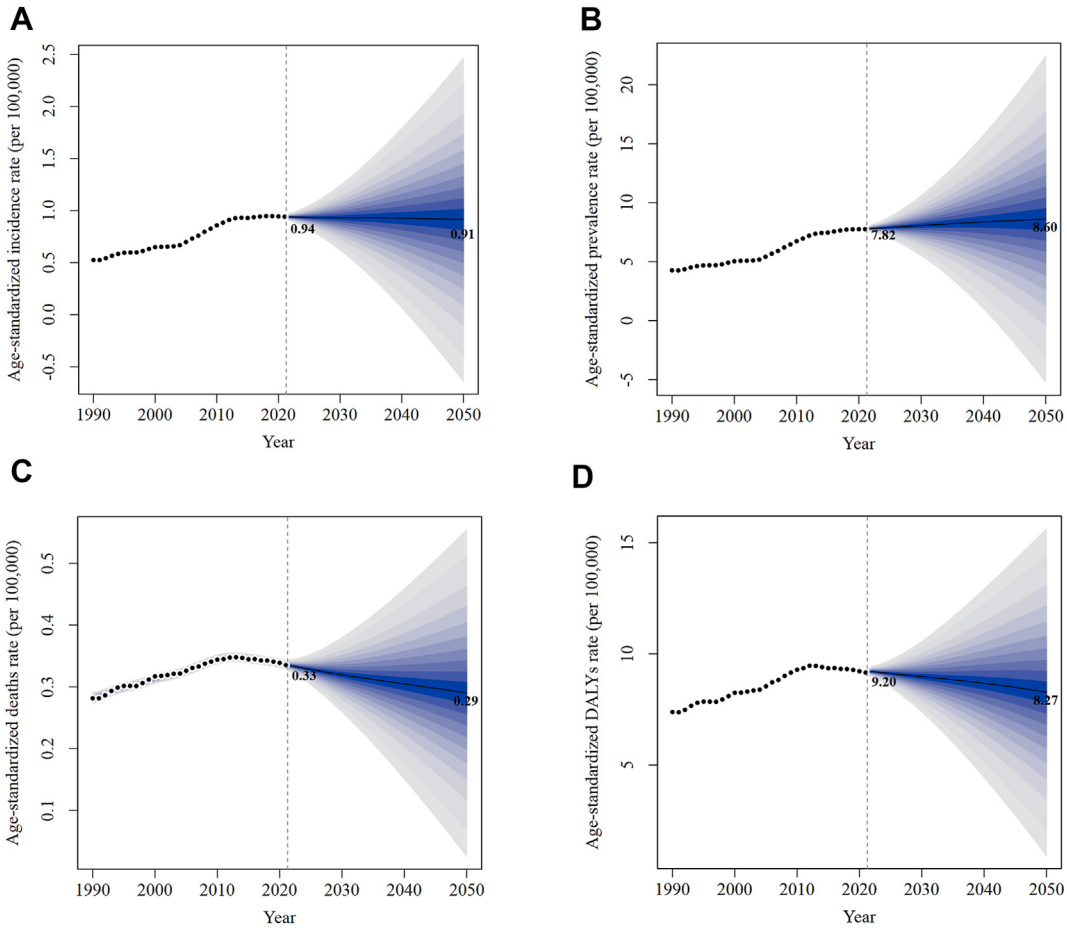


Fig. 5: The forecasts of age-standardized incidence (A), prevalence (B), death (C), and DALYs (D) rates of male breast cancer to 2050. The blue region shows the upper and lower limits of the 95% uncertainty interval (UI).

prevalence is expected to experience the greatest decline in France (–7.97). The largest reduction in the age-standardized rate of death is projected to decrease the most in Lesotho (–0.71), and DALYs are projected to decrease the most in Mozambique (–24.83) (Supplementary Fig. S3 and Table S9).

Risk factors associated with developing MBC

Fig. 6 shows the proportion of MBC deaths and DALYs attributable to the risk factors in 2021. Globally, dietary risks were the leading risk factor for MBC deaths (11.69%, ranging from 6.74% to 13.67% among the 21 regions) and DALYs (11.74%, ranging from 6.78% to 13.67% among the 21 regions) in 2021. Alcohol use contributed to 6.97% of MBC deaths and 7.62% of DALYs worldwide. The proportion of MBC deaths due to alcohol use was highest in Western Europe, followed by Australasia, Central Europe, and Southern Latin America, whereas the highest proportion of alcohol use-attributable DALYs was in Australasia, followed by Western Europe, Central Europe, and Eastern Europe (Fig. 6). Tobacco led to a smaller proportion of MBC deaths (0.83%) and DALYs (0.85%) worldwide in 2021.

Discussion

This study provides a comprehensive overview of MBC burden over the past three decades worldwide. To our knowledge, it is the first report detailing the global burden of MBC, including incidence, prevalence, mortality, and DALYs, as well as assessing the contribution of various risk factors and projecting future trends, thus offering significant insights for addressing this cancer on a global scale. Our results suggest that the epidemiological features of MBC burden exhibited significant regional variations in age-standardized rates of incidence, prevalence, mortality, and DALYs across countries and territories. Alcohol use, dietary risks, and tobacco were important modifiable contributors to MBC deaths and DALYs, making them crucial targets for decreasing the global burden of MBC in the future.

In 2021, a greater MBC burden regarding incidence, prevalence, mortality and DALYs was found in Eastern Sub-Saharan Africa, which encompassed 14 of the top 20 countries with the highest age-standardized rates of incidence (ranging from 1.80 to 4.54 per 100,000 people) among the 204 countries and territories, namely Uganda, Zambia, Mozambique, Kenya, Malawi, South Sudan, Djibouti, Rwanda, Somalia, Eritrea, Ethiopia, Comoros, and Burundi. Race may contribute to the high incidence in this region, which is mainly inhabited by Black individuals, as Black men have been reported to have a higher MBC incidence risk compared to other races, such as White men, which may be linked to genetic factors, lifestyle choices, and socio-economic determinants of health.[23] To address this, healthcare policies in this region may focus on reducing disparities by providing equitable access to healthcare services, including screenings, and ensuring that these services are culturally and linguistically appropriate for minority populations. On the contrary, Zimbabwe in this region had a relatively low MBC burden. This variation may be attributed to underreporting or delayed diagnosis, potentially caused by severe political, economic, and social crises, as well as a significant unawareness of breast cancer in Zimbabwe.[24,25] In addition, East Asia and High-income North America also had a relatively higher MBC burden in terms of age-standardized rates of incidence and prevalence, while the mortality and DALYs rates in these areas are comparatively low. This may be attributed to advanced healthcare infrastructure that enable early detection and effective treatments,

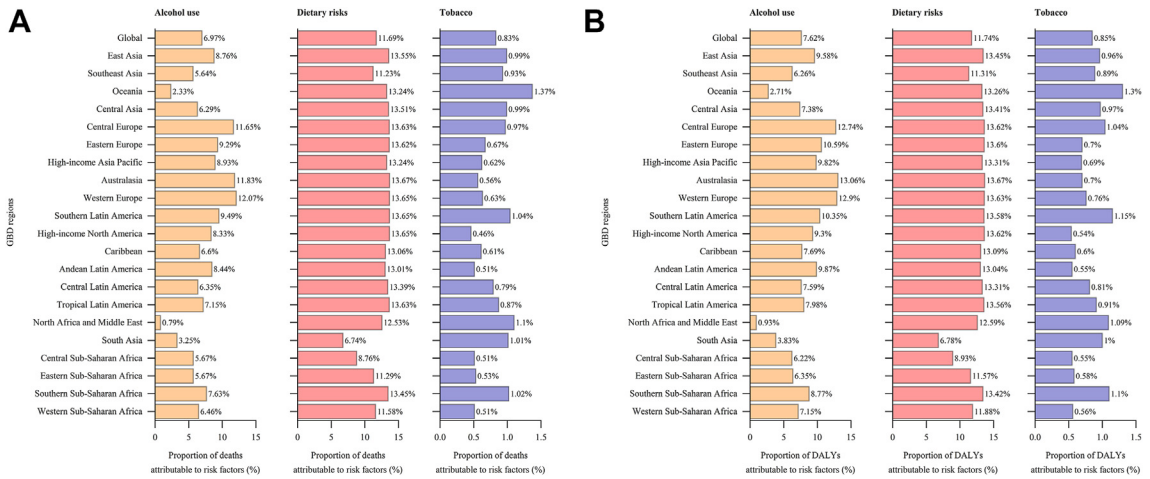


Fig. 6: Proportion of male breast cancer death (A) and DALYs (B) attributable to alcohol use, dietary risks, and tobacco among 21 GBD regions in 2021.

leading to improved survival and reduced disability. Healthier lifestyle habits, such as improved diet and lower alcohol consumption, could also contribute to better outcomes in thes regions.[26,27] While MBC incidence and prevalence are higher, the lower mortality and DALYs highlight the importance of healthcare infrastructure and early intervention in improving patient outcomes.

Previous studies have shown that the overall cancer burden varies across SDI levels, with a strong positive relationship observed between cancer incidence and mortality burden and higher SDI levels.[28,29] However, the age-standardized rate of incidence of MBC was highest in the high-middle SDI quintile, followed by low SDI, middle SDI, high SDI, and low-middle SDI. This pattern may reflect the unique socioeconomic and environmental risk profile of male breast cancer, driven by the combined effects of optimized healthcare resources, the clustering of lifestyle-related risks, and environmental exposures. A deeper understanding of the incidence patterns in these regions will be crucial for developing more targeted prevention and intervention strategies, while also highlighting the complexities of health inequities across different SDI levels. The mortality and DALYs burden of MBC was negatively correlated with SDI levels; higher SDI quintiles had lower death and DALYs rates. As a result of the lowest death rate, regions with high SDI have the highest age-standardized prevalence rates, indicating that MBC patients in high SDI regions may benefit from more structured follow-up systems and broader patient support networks, which are important factors for improving patient survival.[30,31]

From 1990 to 2021, the global MBC burden was significantly increased. We observed that East Asia, Tropical Latin America, Caribbean, and Central Asia were the top four regions that experienced significant increases in age-standardized rates of MBC incidence. The countries included in these four regions are mainly in middle or high-middle SDI quintiles. Several factors may contribute to this increase. First, these regions have undergone demographic transitions, including increasing life expectancy and aging populations, which may contribute to the higher incidence of MBC.[32,33] Moreover, improvements in healthcare infrastructure, diagnostic capabilities, and social awareness have enhanced detection and reporting, uncovering previously underdiagnosed cases rather than a true rise in incidence.[32–34] Additionally, the increase in MBC incidence may also be partly driven by changes in certain risk factors, particularly those influencing estrogen levels either directly or indirectly. These factors include increased alcohol consumption, potential radiation exposure, the growing prevalence of obesity, physical inactivity, poor dietary habits, and hormone therapy use among transgender women.[35–38] However, unlike female breast cancer, which has declined in middle high and high-SDI regions due to successful prevention and screening efforts,[39] MBC incidence in middle or high-middle SDI regions continues to rise. This reflects the complex interaction of demographic, socioeconomic, social cognition, and lifestyle factors.[37,40] Moreover, the age-standardized mortality rates were significantly increased in 10 of the 21 regions from 1990 to 2021, whereas Eastern Europe, High-income Asia Pacific, Australasia, and Western Sub-Saharan Africa experienced significantly decreased of the mortality rates. The decline in mortality and DALYs in these regions may be attributed to advancements in targeted therapies and surgical interventions, which have improved survival rates and reduced disability associated with MBC.[39] Meanwhile, the burden of MBC in Georgia has sharply increased between 1990 and 2021, likely driven by the challenges of rapid population aging—with an estimated one-fifth of the population projected to be aged 65 and older by 2030—and improvements in the healthcare system, which have enhanced the use of modern diagnostic technologies.[41–43]

Over the next three decades, our projections suggest that the global age-standardized rates of incidence, mortality, and DALYs for MBC will decrease, with Western Europe, Southern Sub-Saharan Africa, and Southern Latin America expected to experience the largest reductions in these respective metrics. The decline in MBC incidence in Western Europe over the next 30 years may be attributed to several interrelated factors. First, the advancement of precision medicine has enabled high-risk individuals to undergo earlier genetic screening and tailored risk interventions, particularly those with BRCA1/2 mutations or a strong family history.[44] Additionally, strict regulatory measures on industrial pollutants and endocrine-disrupting chemicals may educe environmental exposures linked to MBC.[45] Previous studies have suggested that rural areas, where industrial pollutants and endocrine-disrupting chemicals may be more prevalent, could experience higher incidences of MBC.[46] In Sub-Saharan Africa, the increase in healthcare funding and the gradual expansion of oncology services are beginning to address historical disparities in cancer care, which likely contribute to the observed reductions in mortality rates in the region.[47] Southern Latin America's improvements may stem from targeted public health measures addressing risk factors like obesity and alcohol consumption, alongside expanded access to cancer treatment.[48–50] These trends highlight the impact of tailored healthcare improvements while emphasizing the need to address persistent inequities to sustain progress. However, the age-standardized incidence rates of MBC in Oceania, Southern Latin America, Central Latin America, South Asia, and Central Sub-Saharan Africa are projected to increase by more than 60% in the next 30 years. Additionally, Eastern Sub-Saharan Africa, the region with the highest MBC burden, is expected to increase by 38.69% by 2050. This highlights

the need for focused attention on improving healthcare access, diagnostic capabilities, and treatment outcomes in these regions. These countries and regions should closely monitor the occurrence of MBC. Considering the difficulty in conducting clinical studies on MBC due to the limited number of cases in a single country, clinical trials can be carried out in these high-incidence areas or countries with a large number of MBC cases in the future, such as China (with 145,449 MBC cases in 2021), the United States (with 33,816 cases), India (with 20,980 cases), and France (with 8213 cases). These studies will be essential to further understand the epidemiology of the disease, identify risk factors, and assess treatment responses, ultimately providing valuable insights for MBC prevention strategies.

Although the global lifetime risk of developing MBC (0.095%) is much lower than that of female breast cancer (ranging from 2.67% to 13.88%),[51] there is significant variation among different regions (ranging from 0.033% in Western Sub-Saharan Africa to 0.179% in Eastern Sub-Saharan Africa) and countries (ranging from 0.002% in Belarus to 0.336% in St. Kitts and Nevis). This huge variation underscores the unbalanced MBC burden distribution worldwide, highlighting the importance of developing country-specific MBC prevention strategies. Our analysis also showed that although the age-standardized incidence rate of MBC in the high SDI quintile was lower than in the low SDI quintile, the high SDI quintile had a higher lifetime risk of developing MBC than the low SDI quintile. The higher lifetime risk in the high SDI quintile may be attributed to the longer life expectancy of individuals in this region[52] since a longer lifespan results in a higher accumulated lifetime risk of developing MBC. From birth, the accumulated risk of developing MBC increased rapidly after the age of 45 years. Consequently, the age-standardized incidence rate of MBC significantly increased from 1.19 per 100,000 people in the age group of 45–49 years to 5.86 per 100,000 people aged 70–74 years. Consistent with the mortality rate, the lifetime risk of dying from MBC was negatively correlated with SDI levels; males in low SDI regions had a much higher lifetime risk of dying from MBC than those in regions with higher SDI levels. Hence, raising public awareness of MBC in countries with low SDI levels is essential to control exposure to major risk factors and facilitate early diagnosis at a lower stage, ultimately leading to a decreased lifetime risk and better survival outcomes for MBC.

Various risk factors have been identified that may contribute to the development of MBC, including modifiable risk factors, such as obesity and physical inactivity, and non-modifiable risk factors, such as age and family history.[40] Our analysis showed that dietary risk was an important risk factor for MBC deaths (11.69%) and DALYs (11.74%) worldwide. Further examination of dietary risks in the GBD revealed that a diet high in red meat was the major dietary risk leading to MBC deaths and DALYs. Unhealthy dietary patterns, particularly the excessive consumption of red meat and processed foods, are linked to metabolic changes and increased inflammation, which may contribute to worse MBC outcomes.[37,53] These dietary habits, often associated with obesity and other chronic conditions, can exacerbate the severity of MBC and lead to higher mortality and disability. Moreover, alcohol use remains an important risk factor for MBC deaths (6.97%) and DALYs (7.62%), especially in Western Europe, Australasia, Central Europe, Southern Latin America, and Eastern Europe. Most of these regions are in high SDI, where the highest average of standard alcohol consumed daily among males is observed.[54] According to an earlier study conducted in Europe, the risk of developing MBC increased by 16% for every 10 g increase in alcohol consumption per day.[55] In addition to being a high-risk factor for MBC, study has reported that alcohol consumption globally contributed to a total of 568,700 cancer cases in males in 2020, with the amount of alcohol consumed being positively associated with the incidence of cancer.[56] In the United States, alcohol accounted for 2.1%–5.0% of all cancer deaths in men.[57] Therefore, the high percentage of MBC deaths and DALYs attributable to alcohol use reinforces the need to strengthen the implementation of alcohol control measures. Tobacco smoking also contributed to the deaths (0.83%) and DALYs (0.85%) of MBC, but its impacts were much less significant than those of dietary risk and alcohol consumption. Thus, avoiding these potential modifiable factors, such as reducing red meat consumption, stopping alcohol consumption, quitting smoking, avoiding radiation exposure, maintaining a healthy weight, and increasing physical activity, may help decrease the risk of developing MBC and mortality from it. Close surveillance with clinical and self-breast examinations is recommended among males with non-modifiable risk factors including aging, BRCA mutation, and family history.

Our study has several strengths. Firstly, it provides a comprehensively global assessment of the MBC burden using high-quality, up-to-date data from the GBD 2021, revealing substantial regional variations and providing epidemiological insights into under-studied countries. Additionally, this study goes beyond cross-sectional analysis by projecting MBC trends up to 2050, aiding future healthcare planning and resource allocation, particularly in regions with rising incidence. Furthermore, this study addresses a critical gap by including lifetime risk estimates and evaluating the impact of modifiable risk factors—dietary risks, alcohol use, and tobacco—on MBC deaths and DALYs, offering valuable guidance for public health strategies and prevention efforts. Nevertheless, several limitations need to be clarified. First, the limited diagnostic capabilities and absence of population-based cancer registries in some

developing countries may lead to the underdiagnosis or underreporting of MBC in these regions. Although the GBD estimates attempt to address these gaps through modeling,[28] the data accuracy may still have some limitations in these countries. Second, our estimates and projections do not account for the potential impact of the COVID-19 pandemic on MBC burden between 2022 and 2023, as these data were not available at the time of this analysis. Therefore, our forecasting results should be interpreted independently of the pandemic's influence. Third, our lifetime risk estimates were based on the current cross-sectional data; however, changes in SDI and certain lifestyle risk factors in some countries may impact the subsequent lifetime risk of developing MBC in the future. Fourth, the risk factor analysis for developing MBC was unavailable in the GBD 2021 and only limited behavioral risk factors, such as alcohol use, dietary risks, and tobacco were available for the analysis of MBC-related deaths and DALYs. Other potentially important factors, including estrogen, environmental exposures, and genetic risk factors—on which most MBC-specific prevention strategies rely—are not included in the GBD study, which could be valuable for further research on MBC. Finally, comorbidities are critical factors contributing to MBC-related deaths, but the GBD 2021 does not provide data on this aspect.[13,19] Future research should address the broader disease burden in MBC patients, including mortality related to comorbidities, to provide a more comprehensive understanding of the impact of MBC on overall health.

Despite the rarity of MBC and the decreasing trend in age-standardized mortality rates worldwide, disparities in the MBC burden were evident across geographic regions and the SDI spectrum in 2021, with a greater burden observed in Eastern Sub-Saharan Africa. High SDI regions had the highest lifetime risk of developing MBC, while low SDI region had the highest risk of dying from MBC. These variations may arise from differences in race, diversity of lifestyle habits, exposure to various risk factors, and disparities in medical standards among countries with uneven income levels. Efforts aimed at reducing the MBC burden through improvements in early diagnosis, access to treatment, and the avoidance of modifiable risk factors should take into account the inequities in the global distribution of MBC. Addressing dietary risks, alcohol consumption, and tobacco use are feasible and promising strategies for preventing MBC and alleviating the disease burden, particularly in regions with a high prevalence of MBC. Furthermore, educating men about the risk of breast cancer and raising awareness that they, too, can develop MBC is crucial for enhancing preventive strategies. This can help break the stigma surrounding MBC in men and promote earlier detection and intervention.

Contributors

Study design: FL, GSR, and YHL. Data collection: YH, YHL, and HBH. Data analyses: YH, YHL, HBH, TTW. Results interpretations: All authors. Manuscript writing: YHL, YH. Manuscript proofing: FL, GSR. YH and YHL had access to and verified the underlying data. All authors read and approved the final version of the manuscript.

Data sharing statement

The original data used in the present study can be accessed at the 2021 Global Burden of Disease (GBD) study (https://vizhub.healthdata.org/gbd-results/). The analyzed datasets during the current study are available from the corresponding author on reasonable request.

Editor note

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Declaration of interests

The authors declare that they have no competing interests.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.eclinm.2024.103027.

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